SUMMER ENGINEERING PROJECT • 2026

Ramjet and Scramjet Simulation Concepts

One-dimensional propulsion-cycle models and design-space exploration

PROPULSION	THERMODYNAMICS	SIMULATION

Project overview

A conceptual simulation study for air-breathing high-speed propulsion. The project organizes inlet compression, heat addition, combustor constraints, and nozzle expansion into traceable one-dimensional models for ramjet and scramjet regimes.

Engineering problem

High-speed propulsion performance depends on tightly coupled compressible-flow processes, and ideal-cycle results can be misleading unless losses, thermal limits, and operating assumptions are explicit.

System architecture

- Atmosphere and flight condition define freestream stagnation properties.
- Inlet/diffuser model applies compression with configurable pressure recovery.
- Combustor model applies heat addition, fuel limits, and ramjet/scramjet Mach assumptions.
- Nozzle model estimates expansion, exit state, thrust, and specific impulse trends.

Engineering focus

- Track static and stagnation properties station by station with unit checks.
- Compare ideal and loss-inclusive cases across Mach number and equivalence ratio.
- Treat dissociation, finite-rate chemistry, and multidimensional shocks as future fidelity.

Verification plan

- Recover limiting textbook cases before exploring the design space.
- Check conservation of mass, energy, and physically admissible temperatures and pressures.
- Use sensitivity plots to distinguish robust trends from assumption-driven conclusions.

Status and evidence boundary
Summer 2026 simulation concept; this brief presents model structure, not validated engine performance.